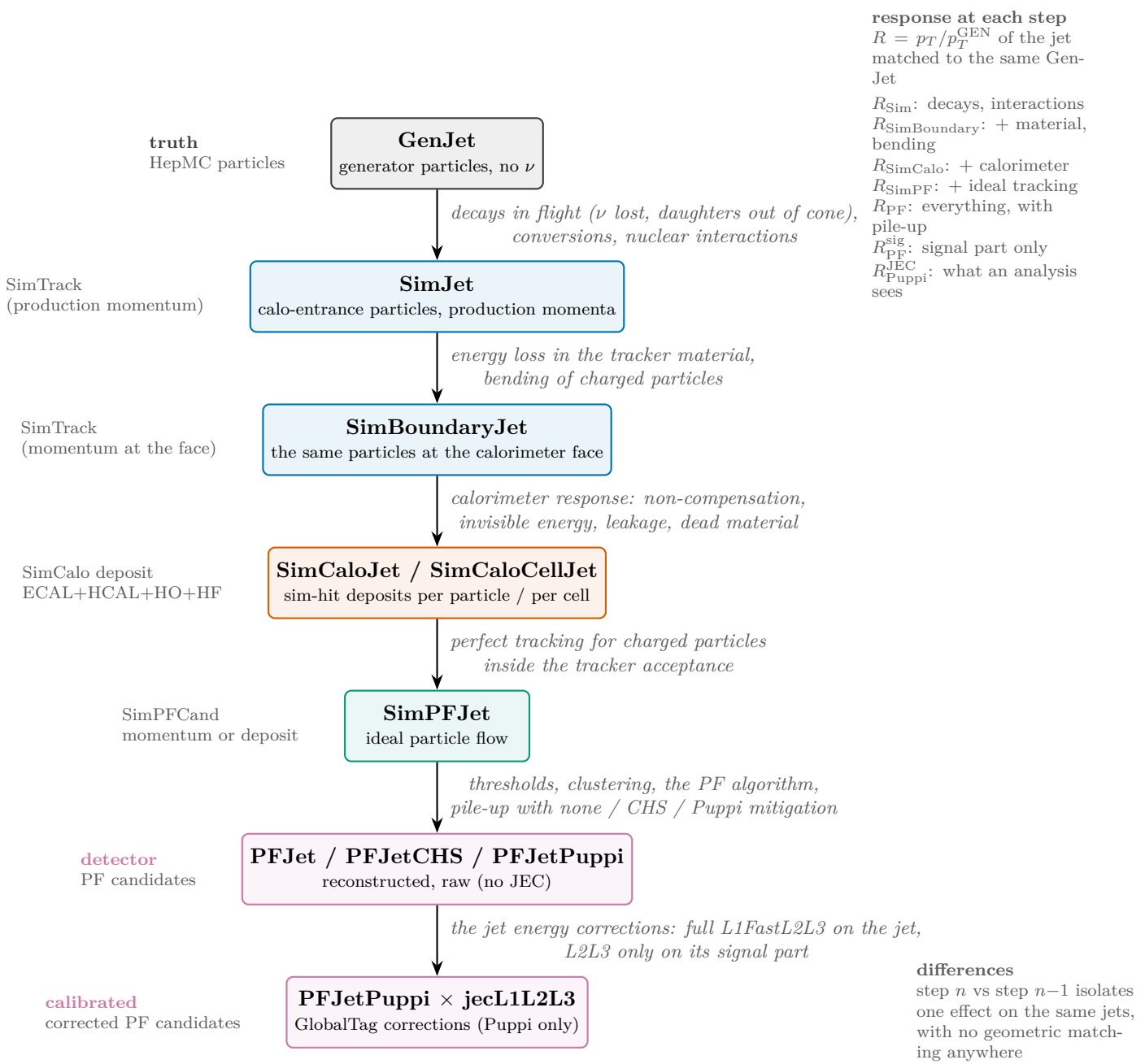

response at each step
$R = p_T/p_T^{\mathrm{GEN}}$ of the jet matched to the same Gen-Jet
R_{Sim}: decays, interactions
$R_{\mathrm{SimBoundary}}$: + material, bending
R_{SimCalo}: + calorimeter
R_{SimPF}: + ideal tracking
R_{PF}: everything, with pile-up
$R_{\mathrm{PF}}^{\mathrm{sig}}$: signal part only
$R_{\mathrm{Puppi}}^{\mathrm{JEC}}$: what an analysis sees
truth
HepMC particles
GenJet
generator particles, no ν
decays in flight (ν lost, daughters out of cone), conversions, nuclear interactions
SimTrack
(production momentum)
SimJet
calo-entrance particles, production momenta
energy loss in the tracker material, bending of charged particles
SimTrack
(momentum at the face)
SimBoundaryJet
the same particles at the calorimeter face
calorimeter response: non-compensation, invisible energy, leakage, dead material
SimCalo deposit
ECAL+HCAL+HO+HF
SimCaloJet / SimCaloCellJet
sim-hit deposits per particle / per cell
perfect tracking for charged particles inside the tracker acceptance
SimPFCand
momentum or deposit
SimPFJet
ideal particle flow
thresholds, clustering, the PF algorithm, pile-up with none / CHS / Puppi mitigation
detector
PF candidates
PFJet / PFJetCHS / PFJetPuppi
reconstructed, raw (no JEC)
the jet energy corrections: full L1FastL2L3 on the jet, L2L3 only on its signal part
calibrated
corrected PF candidates
PFJetPuppi × jecL1L2L3
GlobalTag corrections (Puppi only)
differences
step n vs step n−1 isolates one effect on the same jets, with no geometric matching anywhere